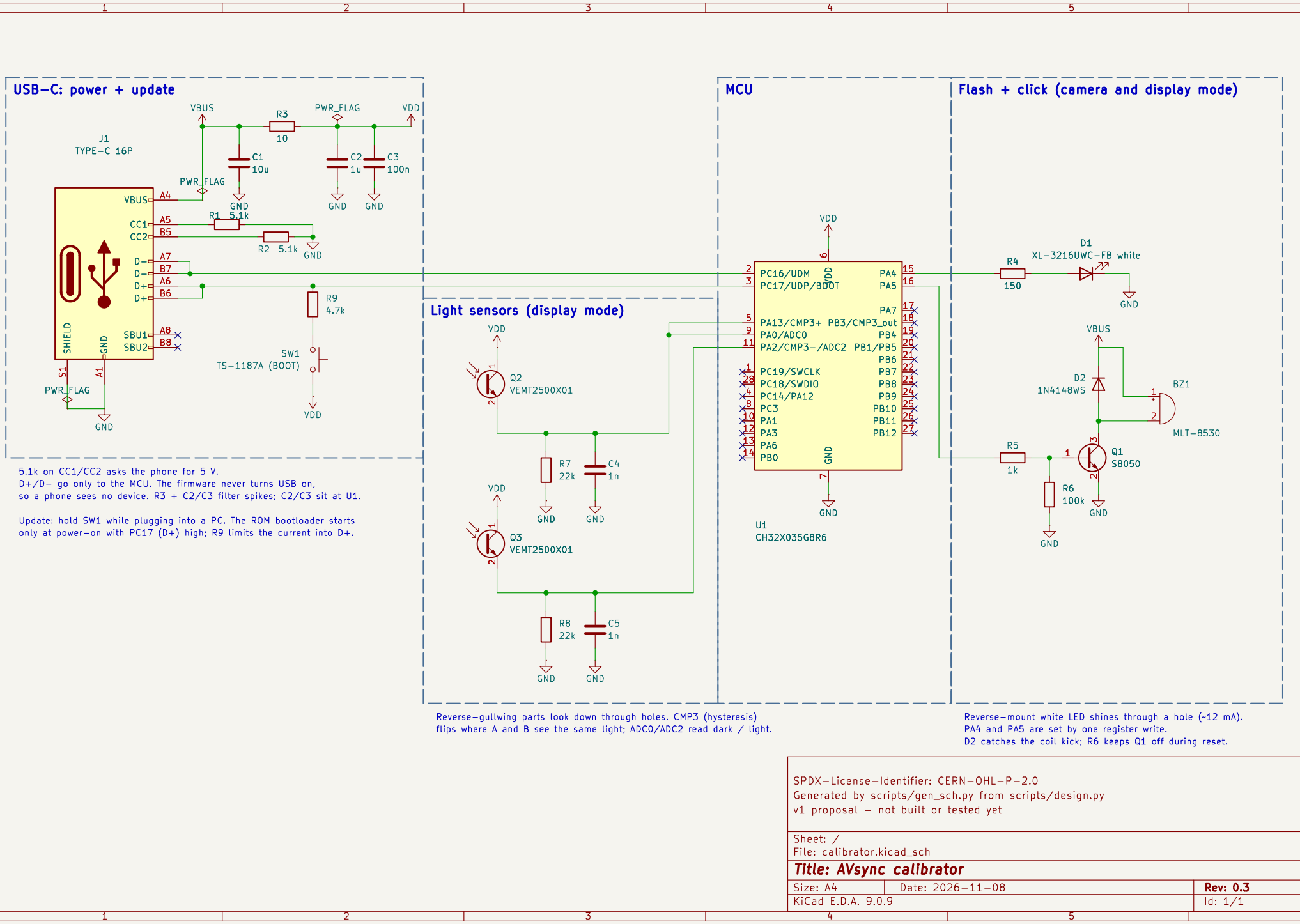

USB-C: power + update
J1
TYPE-C 16P
VBUS
R3
10
PWR_FLAG
VDD
C1
10u
C2
1u
C3
100n
GND
R1 5.1k
R2 5.1k
CC1
CC2
D-
D+
SHIELD
SBU1
SBU2
S1
A1
R9
4.7k
SW1
TS-1187A (BOOT)
5.1k on CC1/CC2 asks the phone for 5 V.
D+/D- go only to the MCU. The firmware never turns USB on,
so a phone sees no device. R3 + C2/C3 filter spikes; C2/C3 sit at U1.
Update: hold SW1 while plugging into a PC. The ROM bootloader starts
only at power-on with PC17 (D+) high; R9 limits the current into D+.
Light sensors (display mode)
Q2
VEMT2500X01
Q3
R7
22k
C4
1n
R8
C5
Reverse-gullwing parts look down through holes. CMP3 (hysteresis)
flips where A and B see the same light; ADC0/ADC2 read dark / light.
MCU
PC16/UDM
PC17/UDP/BOOT
PA13/CMP3+
PB3/CMP3_out
PA0/ADC0
PA2/CMP3-/ADC2
PB1/PB5
PC19/SWCLK
PC18/SWDIO
PC14/PA12
PC3
PA1
PA3
PA6
PB0
PA4
PA5
PA7
PB4
PB6
PB7
PB8
PB9
PB10
PB11
PB12
U1
CH32X035G8R6
Flash + click (camera and display mode)
R4
150
D1
XL-3216UWC-FB white
D2
1N4148WS
BZ1
MLT-8530
R5
1k
Q1
S8050
R6
100k
Reverse-mount white LED shines through a hole (~12 mA).
PA4 and PA5 are set by one register write.
D2 catches the coil kick; R6 keeps Q1 off during reset.
SPDX-License-Identifier: CERN-OHL-P-2.0
Generated by scripts/gen_sch.py from scripts/design.py
v1 proposal — not built or tested yet
Sheet: /
File: calibrator.kicad_sch
Title: AVsync calibrator
Size: A4
Date: 2026-11-08
Rev: 0.3
KiCad E.D.A. 9.0.9
Id: 1/1